Homework 5

Due Date: September 5, 2019 by 5:00pm

MAT 022A Linear Algebra

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Instructions: Write solutions neatly and show all work. No need to show all steps when row reducing. Box your solutions, and explain your answers if necessary. Upload a PDF file of your assignment to Gradescope. When uploading to Gradescope, make sure to select which pages correspond to which question.

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1. (Hill 8.1.12-15) Find the characteristic polynomial, eigenvalues, and eigenvectors of each matrix.

a. $\begin{bmatrix} 2 & -2 & 3 \\ 0 & 3 & -2 \\ 0 & -1 & 2 \end{bmatrix}$

b. $\begin{bmatrix} 2 & 2 & 3 \\ 1 & 2 & 1 \\ 2 & -2 & 1 \end{bmatrix}$

c. $\begin{bmatrix} 2 & 0 & 0 \\ 3 & -1 & 0 \\ 0 & 4 & 3 \end{bmatrix}$

d. $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & 3 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 2 \end{bmatrix}$

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2. (Hill 8.1.16) Find the characteristic polynomial, the eigenvalues and associated eigenvectors of each of the following matrices.

a. $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

b. $\begin{bmatrix} -2 & -4 & -8 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

c. $\begin{bmatrix} 2-i & 2i & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

d. $\begin{bmatrix} 5 & 2 \\ -1 & 3 \end{bmatrix}$

3. (Hill 8.1.T.8) Let λ be an eigenvalue of the nonsingular matrix A with associated eigenvector $\boldsymbol{x}$. Show that $1/\lambda$ is an eigenvalue of A^{-1} with associated eigenvector $\boldsymbol{x}$.

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4. (Hill 8.2.46) Let $A = \begin{bmatrix} 3 & -5 \\ 1 & -3 \end{bmatrix}$. Compute A^9.

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5. (Hill 6.8.2) Which of the following are orthonormal sets of vectors?

a. $\left\{ \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3} \right), \left(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3} \right), \left(\frac{2}{3}, -\frac{2}{3}, \frac{1}{3} \right) \right\}$

b. $\left\{ \left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}} \right), \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right), (0, 1, 0) \right\}$

c. $\{(0, 2, 2, 1), (1, 1, -2, 2), (0, -2, 1, 2)\}$

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6. (Strang 4.4.6) If Q_1 and Q_2 are orthogonal matrices, show that their product $Q_1 Q_2$ is also an orthogonal matrix. (Use $Q^T Q = I$.)

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7. (Hill 6.8.8) Use the Gram-Schmidt process to find an orthonormal basis for the suspace of $\mathbb{R}^4$ with basis $\{(1, 1, -1, 0), (0, 2, 0, 1), (-1, 0, 0, 1)\}$.

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8. (Hill 6.8.12) Use the Gram-Schmidt process to construct an orthonormal basis for the subspace W of $\mathbb{R}^3$ *spanned* by $\{(1, 1, 1), (2, 2, 2), (0, 0, 1), (1, 2, 3)\}$.

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9. (Hill 6.9.2) Let W be a subspace of $\mathbb{R}^4$ with basis $\{\boldsymbol{w}_1, \boldsymbol{w}_2\}$, where $\{\boldsymbol{w}_1 = (1, 1, 1, 0)\}$ and $\{\boldsymbol{w}_2 = (1, 0, 0, 0)\}$. Express $\boldsymbol{v} = (1, 1, 0, 0)$ as a sum of a vector $\boldsymbol{w}$ in W and a vector $\boldsymbol{u}$ in $W^{\perp}$.

10. (Hill 7.1.4-6) Compute the QR-factorization of A.

a. $A = \begin{bmatrix} 2 & -1 \\ -1 & 3 \\ 0 & 1 \end{bmatrix}$

b. $A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 2 & 0 \\ -1 & -2 & 2 \end{bmatrix}$

c. $A = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 2 & -2 \\ 0 & 1 & -2 \end{bmatrix}$

11. (Hill 6.9.T.1) Show that if W is a subspace of $\mathbb{R}^n$, then the zero vector of $\mathbb{R}^n$ belongs to $W^{\perp}$.